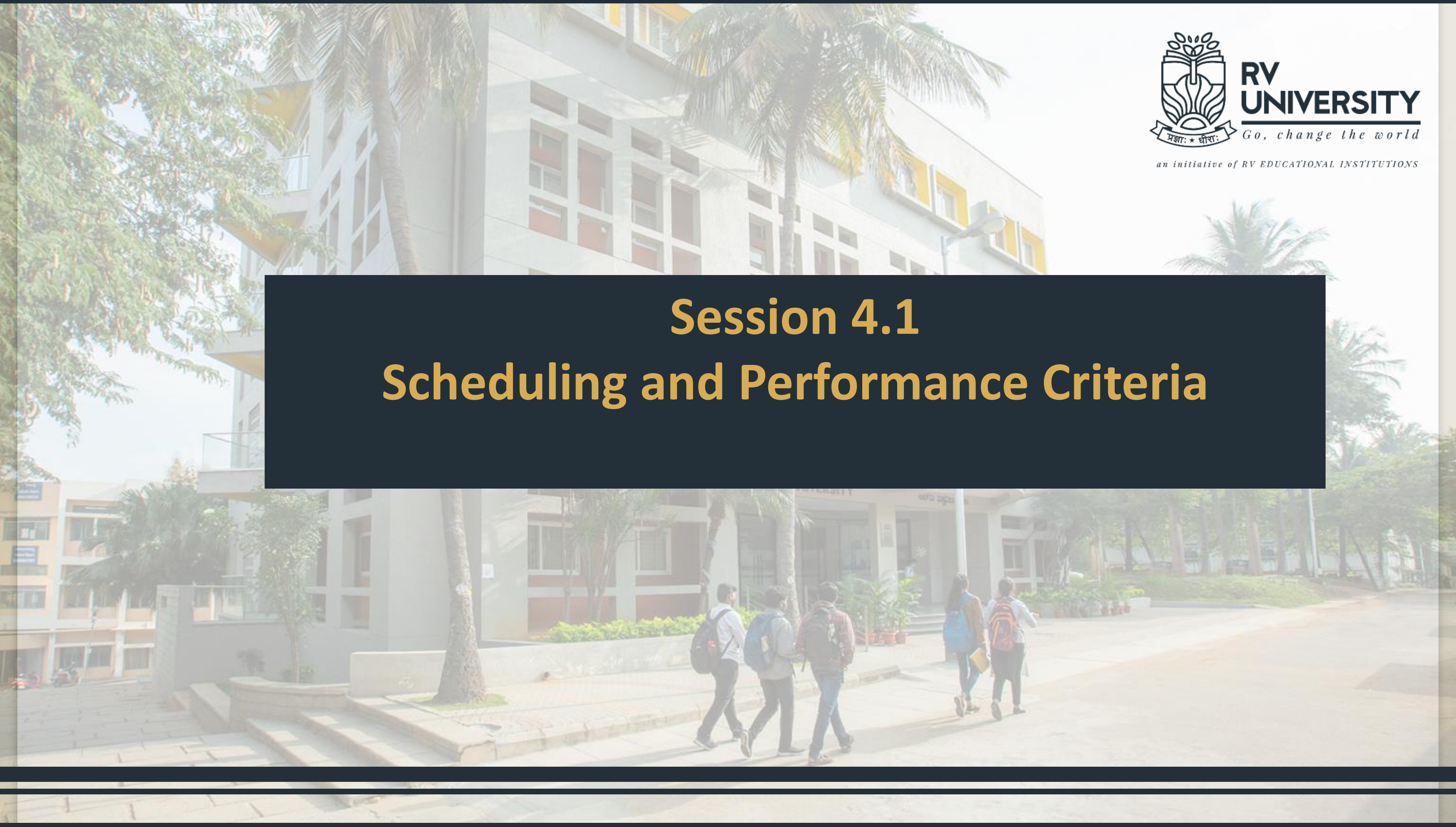

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Session 4.1
Scheduling and Performance Criteria

Session 4.1: Focus

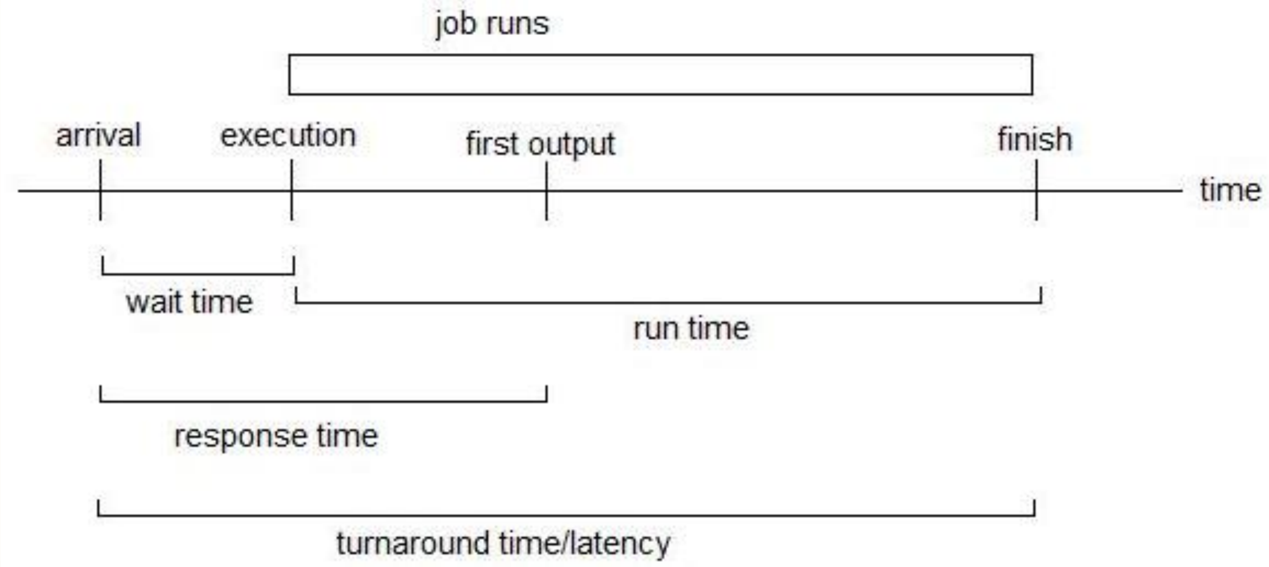
- Preemptive and non-preemptive Schedulers
- Scheduling and Performance Criteria
  - Throughput
  - CPU Utilization
  - Turnaround time
  - Waiting time
- Example problems
- Quiz 1 to 4

Two types of Scheduling
and
Performance Criteria

Two types of Scheduling

- Scheduling in which a running process (on CPU) can be interrupted for one of the following reasons:
  - If a higher priority process enters the queue or
  - The time quantum is over for the currently running process, and it is forcibly taken out of CPU is called **preemptive scheduling**.
  - In this case, the currently running process is moved to the ready queue, and the process at the head of Ready queue or a higher priority process is made to run on the CPU.
- The scheduling in which a running process on the CPU, cannot be interrupted by the OS is called **non-preemptive scheduling**.
  - Any other process which enters the Ready queue has to wait until the currently running process finishes its execution on the CPU and voluntarily relinquishes the CPU.
  - There is no fixed time quantum is allocated to the running processes, it is free to occupy the CPU until it relinquishes the CPU on it own.

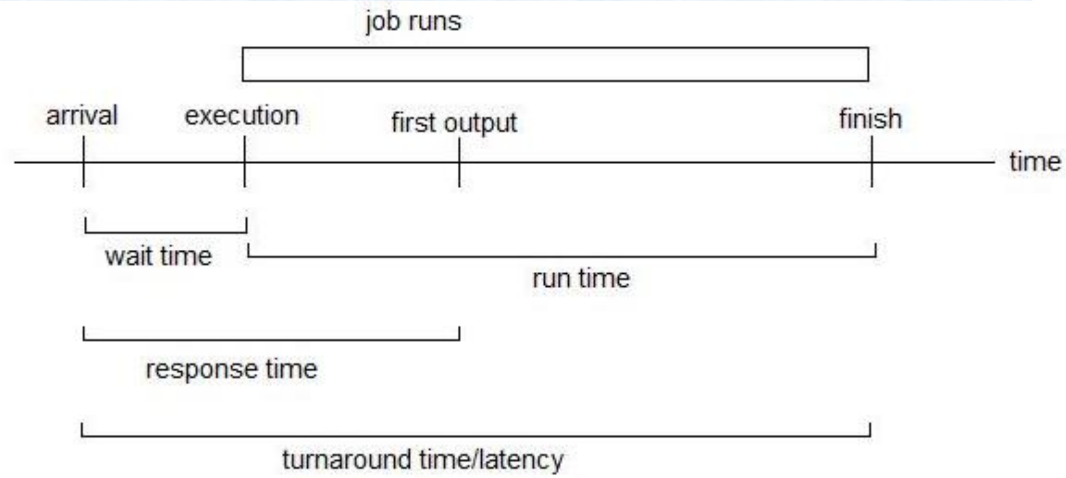
Scheduler Performance Related Parameters



- Scheduler performance is measured in terms of various timing parameters associated with processes or jobs being scheduled by the Scheduler.
- The goal of the scheduler is to make the tasks to complete their execution at the earliest by utilizing the CPU efficiently.

Scheduling Criteria

- **CPU utilization** – Keep the CPU as busy as possible productively.
- **Throughput** – No. of processes that complete their execution per time unit.



- **Turnaround time** or **Latency** – Total amount of time taken to execute and complete a particular process, from the time the process initially became ready.
- **Waiting time** – Total amount of time a process has been waiting in the ready queue.
- **Response time** – Amount of time taken from the time when a job request was submitted (job released or became ready) till the time at which the first response it produced (for interactive jobs).

Quiz 1: Optimization Criteria

- The scheduling criteria are optimization problems.
- Scheduler should either maximize or minimize each of them.
- **Question**: **Maximize** or **Minimize**?
  - CPU Utilization: ↑
  - Throughput: ↑
  - Turnaround time: ↓
  - Waiting time: ↓
  - Response time: ↓
- Can all criteria be optimized simultaneously?
- Usually scheduling algorithms try to optimize average values instead of individual values.

Maximize ↑

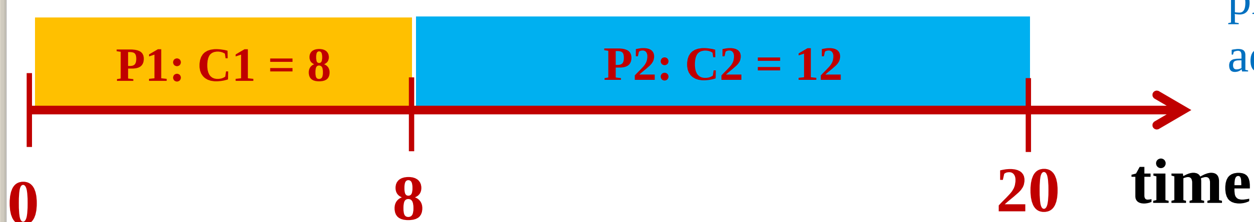
Minimize ↓

Example 1: Compute Scheduling Criteria

- Suppose processes P1 and P2 arrive at time **zero.**
- P1: C1 = 8 and P2: C2 = 12 (computation time for each).
- Consider the time taken for the dispatch of each process is negligible.
- Assume **non-preemptive** scheduler is running.
- Compute the following:
  - Throughput: $\frac{\textbf{No. of Processes completed}}{\textbf{total time}}$ **= 2/20 = 0.1**
  - CPU utilization: **= 100%**
  - Avg. Turnaround: **= (8 + 20)/2 = 14**
  - Avg. Wait time: **= (0 + 8)/2 = 4**

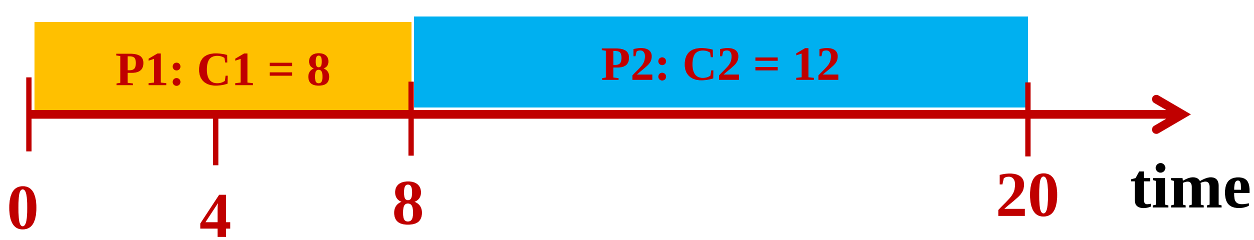
Note 1: Assume time taken by the Scheduler for context switch is negligible,
Note 2: When two processes arrive at the same time, process with lower ID is added to the queue first.

Note 3: CPU utilization is 100% because time taken by the scheduler is negligible.



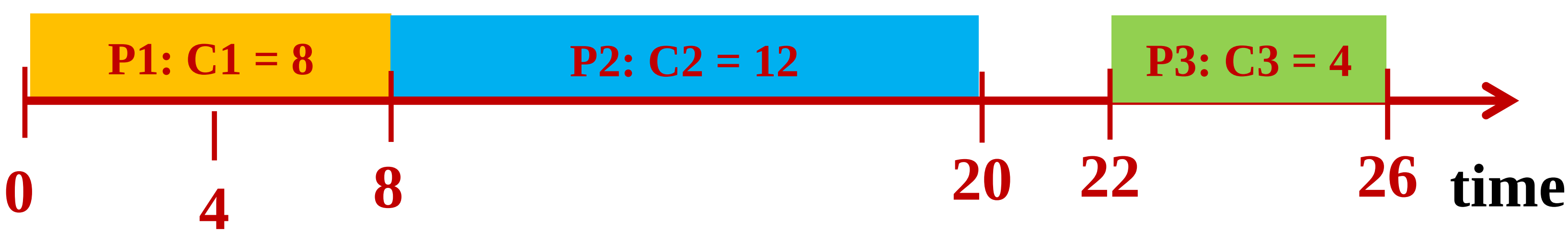
Example 2: Computing Criteria

- Suppose process **P1** arrives at time **zero** and **P2** at **4**
- P1: C1 = 8   P2: C2 = 12 (computation time)
- Consider the time taken for the dispatch of each process is negligible
- Assume **non-preemptive** scheduling
- Compute the following:
  - Throughput: $\frac{\textbf{No. of Processes completed}}{\textbf{total time}}$ **= 2/20 = 0.1**
  - CPU utilization: **= 100%**
  - Avg. Turnaround: **= (8 + 16)/2 = 12**
  - Avg. Wait time: **= (0 + 4)/2 = 2**

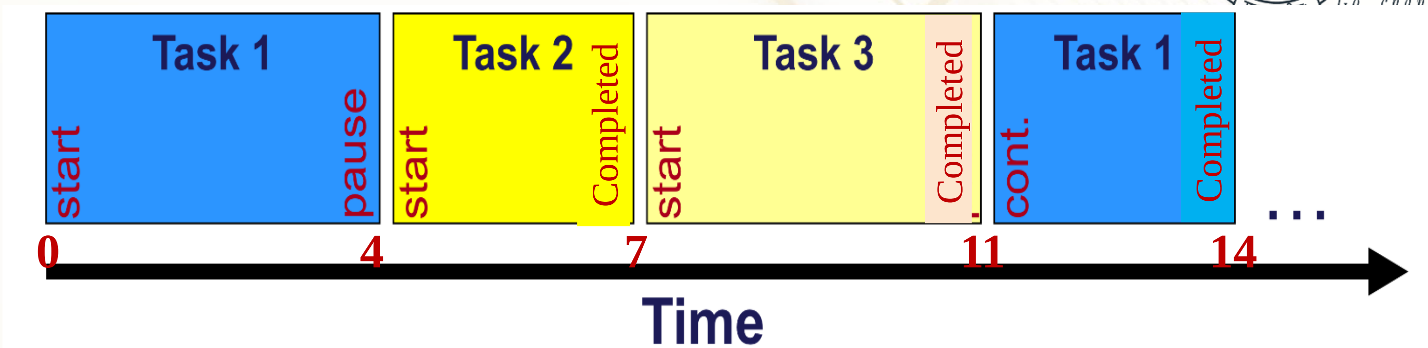


Quiz 2: Computing Criteria

- Suppose process **P1** arrives at time **zero**, **P2** at **4**, **P3** at **22**.
- P1: C1 = 8, P2: C2 = 12, P3: C3: 4 (computation time)
- Assume **non-preemptive** scheduling
- Compute the following:
  - Throughput: $\frac{\text{No. of Processes completed}}{\text{total time}} = 3/26 = 0.115$
  - CPU utilization: **= 24/26 = 92.3%**
  - Avg. Turnaround: **= (8 + 16 + 4)/3 = 28/3 = 9.33**
  - Avg. Wait time: **= (0 + 4 + 0)/3 = 4/3 = 1.33**



Quiz 3: Choose all the correct options



Note: Assume all the tasks given above get completed. They have nothing to execute beyond what is shown here.

- Choose all the correct statements about the tasks given above:

A. Execution time of Task 3 is the largest of all.

ANS: B and C

B. The task with the shortest execution time is Task 2.

C. The task with the largest execution time is Task 1.

D. All the tasks here got completed without any context switchs in between

E. Certainly Task 3 should have become ready after Task 1 and Task 2.

Note: Though Task 3 is taken for execution later, nothing can be said about its time of becoming ready until specifically mentioned.

Quiz 4: PCB entries

- Assume **Process 1** is moved from **running state to blocked state**, because of an I/O request then choose the elements in it's PCB which would have **got changed certainly** because of this state change and give a reason for each:

ANS:

1. State field changes from Running to Blocked.

2. Program Counter value stored here is the address of the instruction to be executed when this process gets a chance to execute next.

3. Context data will be changed fully.

4. I/O status information gets changed because the state change is from Running to Blocked, because of this process having initiated an I/O request.

5. Accounting information, if it contains for example, I/O devices accessed, it would have got changed certainly.

Identifier
State
Priority
Program Counter
Memory Pointers
Context Data
I/O Status Information
Accounting Information
⋮

PCB of Process 1

Note: Process Identifier will remain the same throughout the life of a process.

Session 4.1: Summary

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- Scheduling and Performance Criteria
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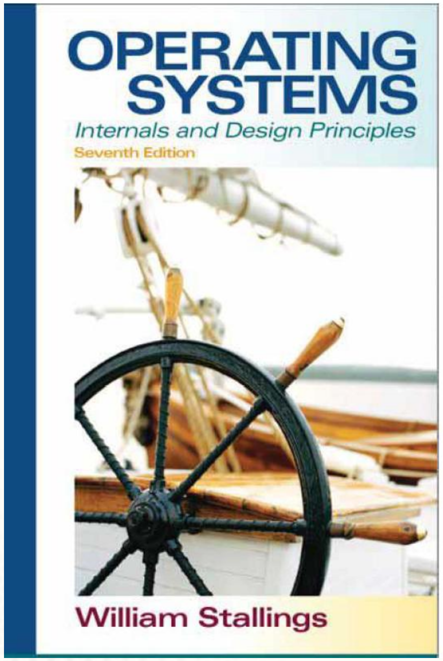
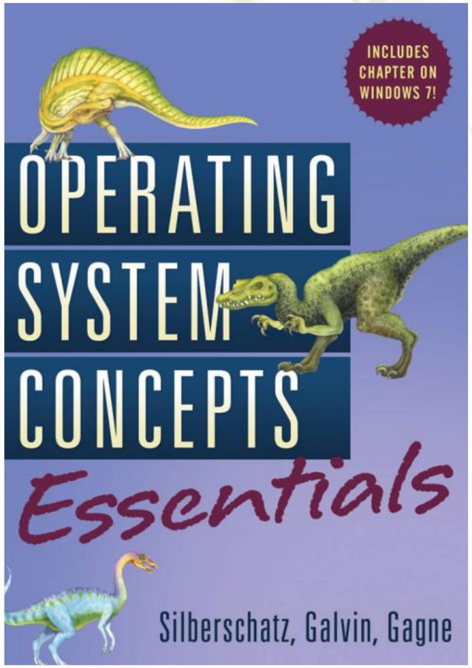
References (1 to 5)

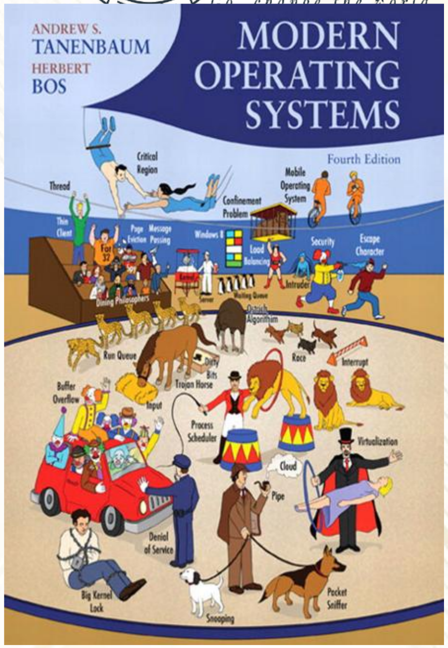
Ref 1



Ref 2



Ref 3



Ref 4

RP2040 A microcontroller by Raspberry Pi

RP2040 Datasheet
A microcontroller
by Raspberry Pi

Ref 5

RP2040 A microcontroller by Raspberry Pi

Raspberry Pi Pico C/C++ SDK
Libraries and tools for
C/C++ development on
RP2040 microcontrollers